



# GetSnap

## Instrument Settings Guide — Adding a New Instrument Yourself

No programming required. Just fill in a form to add support for a new instrument.  
This guide walks through the whole process using a YOKOGAWA instrument as the example.  
The same steps work for any manufacturer.

# Instrument Settings Guide

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GetSnap keeps track of, for each type of instrument you connect, "what commands to send to capture the screen as an image" — this is called an **instrument preset**. If your instrument isn't in the list yet, **you don't need to edit source code or ask a developer**. You can add it yourself from the "Instrument Settings" screen in the "Settings" tab, just by filling out a form.

This guide assumes no programming experience. The only genuinely tricky part is "finding the right commands in the instrument's manual" — using GetSnap itself is straightforward.

## 1. What you'll need before starting

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| What you need                                                             | Why                                                                                                                                                 |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| The target instrument's programming manual (SCPI command reference, etc.) | Usually available as a PDF from the manufacturer's website — search by model number. You'll use it to find the "screen save" / "hardcopy" commands. |
| A way to connect the instrument to your PC (USB, GPIB, or LAN)            | You'll need to verify with the real instrument that what you entered actually works.                                                                |
| The instrument itself, powered on                                         | You'll be checking your settings against it as you go.                                                                                              |

## 2. Overview of the process

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There are six steps in total.

|   |                                                           |
|---|-----------------------------------------------------------|
| 1 | Open the "Instrument Settings" screen                     |
| 2 | Click "+ Add New"                                         |
| 3 | Find two pieces of information in the instrument's manual |
| 4 | Fill in the form's required fields                        |

|   |                                               |
|---|-----------------------------------------------|
| 5 | Click "Save"                                  |
| 6 | Connect the real instrument and try a capture |

## STEP 1: Open the "Instrument Settings" screen

Launch GetSnap and open the "Settings" tab at the top. Near the bottom, under the "Instruments" section, click the "Instrument Settings" button.

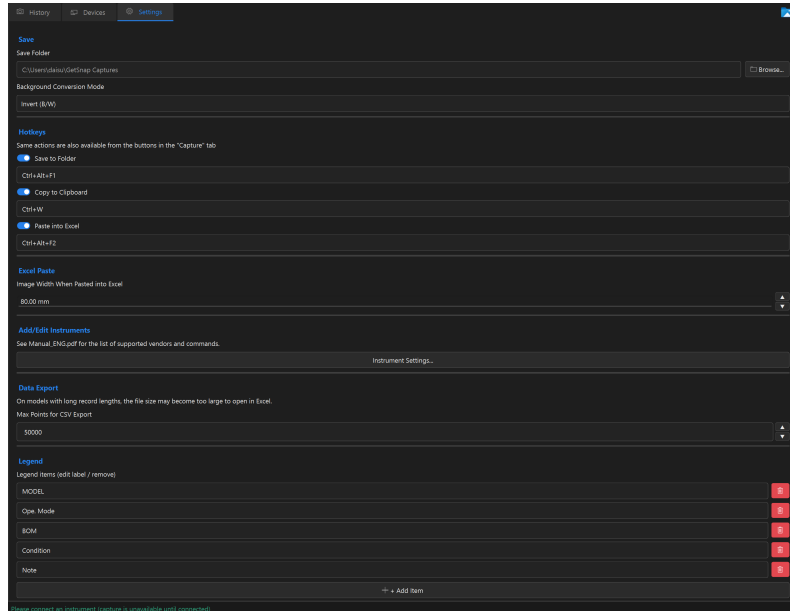


Figure 1: The "Instrument Settings" button in the Settings tab

## STEP 2: Click "+ Add New"

A dialog opens. The left side lists your registered instruments; the right side is the edit form. Click "+ Add New" at the bottom left, and the form on the right clears out, ready for you to register a new instrument.

Registered instruments

Tektronix
Yokogawa
LeCroy
Keysight E5061B

+ Add New
Import
Export

Save
Delete

[Required] Name

[Required] Model match string (substring of the "IDN?" response, matched case-insensitively)

[Required] Image save commands (one per line, last line is the query)

[Required] Commands (SCPI, one per line)

☐ Support no-invert capture

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave this off to reuse the image save commands instead.

Image Format
Delay Between Commands

PNG
0.00 s

☐ Support CSV data export

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled-check command, every channel currently displayed (enabled) on the oscilloscope is automatically fetched and combined into a single CSV.

☐ Support DIV condition display

Used by the "DIV Conditions" feature to auto-reflect the user-selected channels/Math traces' voltage range (V/div) in the image, and the time axis (horizontal) range when "Enable time axis display" is ON in the Settings tab.

[Optional] Notes

Figure 2: The empty form right after clicking "+ Add New"

Fields labeled **[Required]** in red must be filled in before you can save. Fields labeled **[Optional]** can be left blank. Most optional fields are grouped into boxed "feature cards" such as "Support no-invert capture" — the inputs inside only appear once you turn the checkbox on. Leave any feature you don't need turned off.

## STEP 3: Find two pieces of information in the manual

Before filling in the form, look up the following two things in the target instrument's programming manual.

### TO FIND #1

#### Model match string (the \*IDN? response)

Sending the `*IDN?` command to an instrument returns a self-identification string containing the manufacturer name, model number, and so on. Manuals often show an example (e.g. `YOKOGAWA,DLM3054,...`). Use a distinctive part of this string — usually the manufacturer name — as the "model match string."

If you can't find an example in the manual, you can still connect from the "Instrument Connection" tab even without a matching preset — the actual `*IDN?` response is displayed on screen as-is, so you can read it directly from there.

### TO FIND #2

#### The screen-capture (hardcopy) command

Search the manual's table of contents or index for keywords such as:

- `HCOPY` / Hardcopy (printing)
- `Image` / Screen / Display
- `SDUMP` / Dump
- `Print`

What you're looking for is a command that returns the current screen as image data (PNG, BMP, etc.) — a *query*, meaning it ends in `?`. It's common for this to be more than a single command: a setup command or two (choosing the image format, choosing a white/inverted background for

printing) followed by a final query that actually returns the data.

A command that only "saves to a USB drive" or "prints on the instrument's built-in printer" (for example, some instruments' `MMEMory:STORe:SCReen`) can't be used with GetSnap, because the PC never receives the image data directly. Make sure the command you pick is a query that returns the data over the bus.

## STEP 4: Fill in the required fields

**Name**[Required]

The name shown in the list and in capture history. Anything you like (e.g. `Rigol`).

**Model match string**[Required]

Enter the string you found in STEP 3, all in lowercase (matching is case-insensitive either way).

**Image save commands**[Required]

Enter the command(s) from STEP 3, one per line, in the order they should be sent. The last line must be the query that returns the data.

**Commands (return to local)**[Required field, but may be left blank]

Extra commands sent to return the instrument to front-panel control after a capture. GPIB and USB connections already handle this automatically via VISA's standard GTL mechanism, so this can usually stay blank.

# Worked example: adding a YOKOGAWA instrument

Here, we'll assume YOKOGAWA isn't in the list yet and walk through adding it as a concrete example (in the real, shipped version of GetSnap, a Yokogawa preset is already built in — but to make the process tangible, this example uses the exact same values that preset actually uses).

## Result of STEP 3: what was found in the manual

| Purpose                                                 | Command             |
|---------------------------------------------------------|---------------------|
| Select inverted (white-background, print-friendly) mode | :IMAGe:TONE REVERse |
| Select PNG image format                                 | :IMAGe:FORMat PNG   |
| Query the image data                                    | :IMAGe:SEND?        |

On YOKOGAWA's DLM/DL series, sending these three commands in order returns the screen image with a white background (verified against real hardware; DLM2000 Series Communication Interface User's Manual, IM 710105-17E, section 5.16).

## Result of STEP 4: the filled-in form

Registered instruments

Tektronix

Yokogawa

LeCroy

Keysight E5061B

+ Add New

Import

Export

SaveDelete

[Required] Name

Yokogawa (example)

[Required] Model match string (substring of the "IDN?" response, matched case-insensitively)

yokogawa

[Required] Image save commands (one per line, last line is the query)

:IMAGe:TONE REVERse  
:IMAGe:FORMat PNG  
:IMAGe:SEND?

Run Test

[Required] Commands (SCPI, one per line)

Sent in addition to the VISA-level GTL command when returning the instrument to local control after a capture or data export. GPIB/USB connections already use GTL as the primary mechanism, so this can usually be left blank.

Optional

Support no-invert capture

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave this off to reuse the image save commands instead.

[Optional] Commands (one per line, last line is the query)

:IMAGe:TONE COLOr  
:IMAGe:FORMat PNG  
:IMAGe:SEND?

Image Format

PNG

Delay Between Commands

0.00 s

Support CSV data export

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled-check command, every channel currently displayed (enabled) on the oscilloscope is automatically fetched and combined into a single CSV.

Support DIV condition display

Used by the "DIV Conditions" feature to auto-reflect the user selected channels/Math traces' voltage range (V/div) in the image, and the time axis (horizontal) range when "Enable time axis display" is ON in the Settings tab.

[Optional] Notes

Figure 3: The required fields filled in for the Yokogawa example (with "Support no-invert capture" also turned on)

| Field                      | Value entered                        |
|----------------------------|--------------------------------------|
| Name                       | Yokogawa                             |
| Model match string         | yokogawa                             |
| Image save commands        | All 3 lines below                    |
| Commands (return to local) | Left blank (GTL alone is sufficient) |

- :IMAGe:TONE REVerse
- :IMAGe:FORMat PNG
- :IMAGe:SEND?

In Figure 3, the optional "Support no-invert capture" feature is also turned on, with a non-inverted variant of the commands (below). This is used when "No Invert" is selected as the background mode in the Settings tab; if left blank, the image save commands are reused automatically.

- :IMAGe:TONE COLOr
- :IMAGe:FORMat PNG
- :IMAGe:SEND?

## STEP 5: Save

Click "Save" at the top of the form on the right. The new instrument appears in the list on the left, selected. A confirmation message appears at the bottom of the dialog when it succeeds.

The screenshot shows a software interface for saving instrument settings. On the left, a list of 'Registered instruments' includes Tektronix, Yokogawa, LeCroy, Keysight E5061B, and 'Yokogawa (example)', which is highlighted in blue. On the right, a form with 'Save' and 'Delete' buttons contains several sections: a 'Name' field with 'Yokogawa (example)', a 'Model match string' field with 'yokogawa', and a text area for 'Image save commands' containing ':IMAGE:TONE REVerse', ':IMAGE:FORMat PNG', and ':IMAGE:SEND?'. Below this is a 'Run Test' button. Further down is a section for 'Commands (SCPI, one per line)' with an 'Optional' text area. A 'Support no-invert capture' section is checked, with a note about its use. Below that is an 'Optional' section for 'Commands' with ':IMAGE:TONE COLOR', ':IMAGE:FORMat PNG', and ':IMAGE:SEND?'. The 'Image Format' is set to 'PNG' and 'Delay Between Commands' is '0.00 s'. There are also checkboxes for 'Support CSV data export' and 'Support DIV condition display', each with explanatory text. At the bottom is an 'Optional' 'Notes' text area. A confirmation message at the very bottom reads 'Saved \"Yokogawa (example)\".'

Figure 4: After saving — the new entry appears selected in the list, with a confirmation message at the bottom

Saving under a name that already exists in the list produces an error (this prevents accidentally overwriting something). To edit an existing instrument's settings, select it from the list instead of using "Add New."

## STEP 6: Verify with the real instrument

Switch to the "Instrument Connection" tab and connect to the target instrument. If the name shown on connection matches the name of the preset you just added, the model match string worked correctly. In the "Capture" tab, try "Save to Folder" or "Copy to Clipboard" and confirm the image comes out correctly.

If it doesn't work, go back to "Instrument Settings," review the commands, and repeat STEPs 4–6. Everything you've saved can be edited at any time, so feel free to experiment.

## Testing commands before you save: Run Test

Below both the "Image save commands" field and the "Data commands" field is a "Run Test" button. It sends the commands currently typed into that field to the connected instrument right away and shows you the response (the button is disabled while no instrument is connected). Using it before STEP 5 lets you catch a wrong command immediately, instead of saving first and only finding out in STEP 6.

## Back up before you edit: Export / Import

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The "Export" and "Import" buttons (icon-only) at the top of the list let you save a single preset's content to a JSON file, and load one back later. Before editing a built-in instrument (Tektronix, Yokogawa, etc.), it's strongly recommended to "Export" its current content first. Built-in presets have no "revert to original" button, so once you've overwritten one, the only way back is to "Import" the JSON file you exported beforehand.

# About the optional features

The required fields alone are enough for basic screen capture. The following three feature cards only need to be turned on if and when you actually want to use them — there's no need to fill them in right away.

**Support no-invert capture**

Lets you set a separate set of commands used when "No Invert" is selected as the background mode in the Settings tab. If left off, the image save commands are reused instead.

**Support CSV data export**

Enables the "Save CSV Data" button in the Capture tab, which saves the waveform's numeric data as a CSV file.

**Support DIV condition display**

Enables the "DIV Conditions" feature, which automatically writes each channel's/Math trace's voltage range (V/div) onto the captured image.

Full details for each of these fields are also documented in the main `Manual_ENG.pdf` (sections 8.6–8.7).

# This same process works for any manufacturer

This guide used YOKOGAWA as the example, but these six steps are identical regardless of manufacturer. To add support for Tektronix, Rigol, Siglent, LeCroy, Keysight, or anything else, get that instrument's programming manual, find its model match string and screen-capture command in STEP 3, and follow the same process.

Sometimes different product series from the same manufacturer use different commands. In that case, you can register multiple presets with more specific model match strings (including the series name, e.g. `d1m2000`) so that each series gets its own set of commands.

# Editing or removing what you've added

| What you want to do | How |
|---------------------|-----|
|---------------------|-----|

|                                                                  |                                                                                                                                                                        |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fix the settings for an instrument you added                     | Select it from the list, change the values, click "Save"                                                                                                               |
| Remove an instrument you added                                   | Select it from the list, click the red "Delete" button (after a confirmation popup — this cannot be undone)                                                            |
| Edit one of the built-in instruments (Tektronix, Yokogawa, etc.) | Select it from the list, change the values, click "Save" (only that instrument is overridden with your customization — "Export" it first as a backup)                  |
| Undo your changes to a built-in instrument                       | There's no "revert to original" button. "Import" the JSON file you "Exported" before editing                                                                           |
| Remove a built-in instrument from the list                       | Select it from the list, click the red "Delete" button (removes it from the list; there's no button to bring it back, so "Export" it first if you might want it again) |

## Troubleshooting

| Symptom                                                                        | What to do                                                                                                                                                           |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clicking "Save" produces an error                                              | Check that every [Required] field (Name, Model match string, Image save commands) is filled in.                                                                      |
| "An instrument with this name already exists"                                  | Choose a different name, or select the existing preset from the list to edit it instead of using "Add New."                                                          |
| The instrument connects as "unsupported"                                       | The model match string probably doesn't match the real <code>*IDN?</code> response. Compare it against the response string shown in the "Instrument Connection" tab. |
| It connects fine but the capture fails                                         | Check which step the error message names, and review the corresponding command in "Instrument Settings."                                                             |
| The image captures fine but the background doesn't turn white (doesn't invert) | Check the "Settings" tab's background mode selection against which command set (image save vs. no-invert capture) is configured for it.                              |

## Checklist

**1**Obtained the target instrument's programming manual

**2**Confirmed the model match string (from the \*IDN? response)

**3**Confirmed the screen-capture command (a query that returns image data)

**4**Filled in the required fields via "Instrument Settings" → "+ Add New," then clicked "Save"

**5**Connected the real instrument and confirmed the capture succeeds